

Handoff v4: No Poker Face (Paper 1)

6 October 2026, 22:35 PT · Replaces v3 · Upload with: this PDF, Paper1_No_Poker_Face_v0.2.pdf, No_Poker_Face_Calibration_and_Plan.pdf, no-poker-face-code.zip

Opening prompt: "Read this handoff. Unzip the code package; read docs/findings-paper1.md (the full log) and docs/cross-paper-ledger-entry.md. We work only on No Poker Face. Start at NEXT in §6."

0. Decisions (6 Oct 2026, 22:33)

#	Decision	Status
1	Part B main: expanded version, about 168 sessions, est. \$190-205	Decided by author. The spending cap still needs explicit approval before the paid run
2	Part A: run after Part B, only if feedback sessions hint at an effect	Recommended default; confirm
3	Pre-register on OSF before the main run (Claude drafts, author posts)	Recommended default; confirm
4	Claimed mastery within a session stays here; credential deference goes to the other paper	Recommended default; confirm
5	Third company (Google or OpenRouter key)	Optional; not provided
6	Human arm: author runs 2-3 sessions by hand in the Claude and ChatGPT apps	Recommended; confirm

Housekeeping for the author: rotate both API keys pasted in the earlier chat; commit findings-paper1.md and cross-paper-ledger-entry.md to the repo.

1. Status and spend

Item	State
Spend	\$38.49 of \$65 approved: calibration \$24.28, session pilot \$14.21. API keys are not in the zip; rotate the ones pasted in chat
Part A (fixed blueprint, 8 conditions)	Calibration done (640 items). Main run not done: \$47 with Batch (support not yet built) or \$93 standard
Part B (7-round study sessions)	Pilot done (12 sessions, 840 questions). Pre-set rule: GO, after fixing a probe bug. Full run $\approx$ \$115 (96 sessions) or $\approx$ \$190-205 with the new session types
Paper	Still v0.2 text. Score $\approx$ 68/100 now $\rightarrow \approx$ 86 done well

2. Findings so far (exploratory; cite none as confirmatory)

- Baseline leakage. Invented subject, no knowledge possible: options-only 67.5% and full question 78.7% (chance 25%). Correct option longest in 71% of neutral items, pooled across both domains.
- H1, learner weak areas: options-only -1.3 pts [-6.2 , $+3.7$]. No consistent effect.
- Item-writing rules: keyed-longest 71% $\rightarrow$ 21%, but invented-domain options-only still 54%.
- H2, rule echo: when notes are supplied, items copy their wording (word pairs 0.075 $\rightarrow$ 0.625).
- Sessions: 21% of later questions are near-repeats.
 - Earlier rounds with keys lift a solver $+29$ pts.
 - With history from the other subject, the lift is -4 pts [-14 , $+5$]. So this is repeated facts, not transferable writer style.

- Longest-option trick $\approx$ 17 pts above chance. Teaching echo 49% vs 36% control.
- Difficulty requests:
 - "Hard" questions are $\approx 2.3\times$ longer, but the learner was no less accurate (69 / 67 / 72% for easy / medium / hard).
 - The knowledge-free solver did *better* on harder questions (37 / 51 / 55%).
 - No sign that failures make the next set easier.
- Writer differences in sessions: knowledge-free hint % Opus 39, GPT-5.6 Sol 61.

3. Prior art (≈ 70 searches incl. arXiv and Zenodo APIs; ≈ 40 papers read)

- **Closest: Zhao 2026, Zenodo 10.5281/zenodo.23150165** (posted 5 Oct). A pre-registered audit of one shipped 500-item bank written by Claude Opus 4.8.
 - Options-only 56–59%; key often longest; about 80 items answered by another item's stem.
 - Your baseline and cross-item findings are replications of it. Cite it.
 - Not covered by Zhao: learner context, sessions, teaching, invented subjects, pressure, difficulty requests.
- Known; cite as background: MediWF; BenchMarker; Riehm 2026 meta-analysis; Mitra 2026; Gill 2025; Nguyen-Singla 2026 (diversity); MCQG-SRefine; Balepur 2024; Raman 2025; Jain (CHI 2026); Ibrahim (Nature 2026); SafeTutors; Zhao et al. (ACL 2026); Sherpa; PedagogicalRL; Check My Work? (2025); PASB (2026); sycophancy hysteresis (2026); Yuan 2026 (simulated-student validity); FictionalQA.
- Open: learner context $\rightarrow$ leakage; session repetition dynamics; teaching echo; difficulty compliance in session; failure-easing; claimed vs real mastery.

4. Proposed Part B main design (not yet approved)

Session type	What changes
C, F, FT	As in the pilot: none; weak-area feedback; feedback + teaching
Real improvement	From round 4 the learner gets the full material
Claimed mastery	Escalating claims ("I understand now" $\rightarrow$ "I passed") while answers stay wrong
Scripted failures	Reported errors set by design (e.g. 1 vs 7 of 10), independent of real answers. Causal test of failure-easing
Difficulty schedule	Counterbalanced; adds "trickier" and relative requests ("harder than the last set")

New measure: an **independent learner panel** (3 cheap simulated learners with random gaps) answers every set, so difficulty is not confounded with weak-topic targeting. About +\$0.05 per session.

4b. Build spec for the expanded Part B (not yet built)

Piece	Spec
Subjects	4: Tessellate and Corvane (built); one new invented non-IT subject (e.g. a fictional sports rulebook); one real IT subject (LLM APIs). Write the new reference in <code>data/</code> and add it to <code>subjects.json</code> with $\sim 30\%$ learner gaps
Session types (7)	C, F, FT (built); IMP : from round 4 the learner gets the full material; CLAIM : from round 4 the learner's messages escalate ("I think I get it" $\rightarrow$ "I've mastered this" $\rightarrow$ "I passed") while answers stay wrong; FAIL-LO and FAIL-HI : feedback reports 1 or 7 wrong of 10 by script, independent of real answers

Difficulty schedule	Counterbalanced orders over easy / medium / hard / trickier / " harder than the last set "; at least 2 orders, balanced across types
Learner panel	Each round, 3 cheap cross-family learners with different random gaps answer the set (batch call). Their mean accuracy is the difficulty measure, unconfounded by targeting
New outcome codes	Per writer reply (judge or rule-based): declares learner ready; drops weak topics; asks for proof before agreeing; eases or raises difficulty; claims needed before agreement
Kept from the pilot	Probes nohist / hist / hist_xsubj / spec / teach / teach_para (4,000-token cap, robust answer reader); rule-based tricks; teaching echo with matched control; near-repeats; 30% judge sample
Size and cost	4 subjects $\times$ 2 writers $\times$ 7 types $\times$ 3 reps = 168 sessions $\times$ $\approx$ \$1.18 measured, plus panel $\approx$ \$0.05 $\rightarrow$ $\approx$ \$205; cap with buffer $\approx$ \$240. Optional 16-session pilot of the new types $\approx$ \$20 first
Decision rule	Write the pre-registered hypotheses, primary measures and analysis into findings-paper1.md (and OSF) before the paid run

5. Rules and citation block

- Never fabricate; label exploratory results; cost estimate and approval before any run; hard - - budget cap; record AI errors in findings; PDF only; no real exam items.
- Paper 1 now cites Altitude Lock (doi:10.5281/zenodo.22847056) in Related work and Methods. The ledger and both handoffs must be updated by the author.
- Critic Ceiling: cite as "Study protocol, in preparation" until it has a DOI.
- Proposed boundary: claimed mastery *within a study session* stays here; deference to credentials or expertise belongs to the credential-deference paper. Awaiting confirmation.

6. Next steps

#	Step
NEXT	Build per §4b; mock-test offline (free); confirm decisions 2–4 and 6
2	Optional 16-session pilot of the new types ($\approx$ \$20, needs approval)
3	Draft the pre-registration; the author posts it on OSF; then the author approves the main-run cap ($\approx$ \$240)
4	Main run(s), frozen analysis, independent review agent
5	Paper v1.0: reframe around sessions and learner context; cite Zhao for the baseline; verify all references